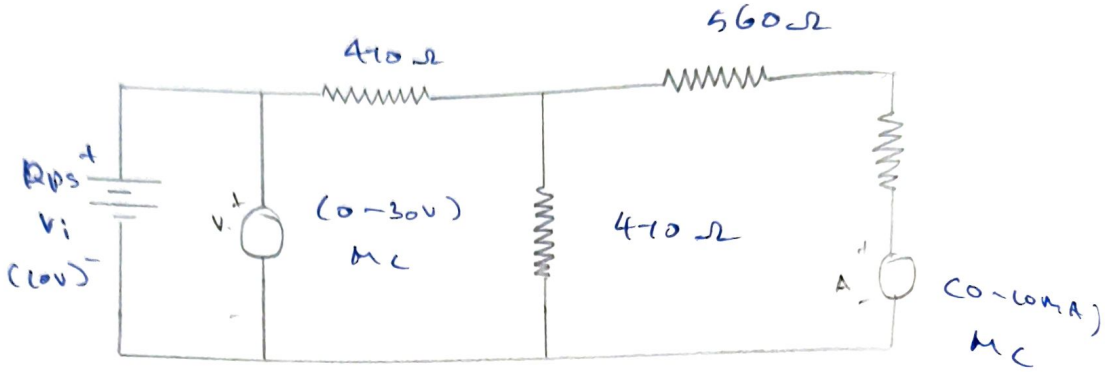
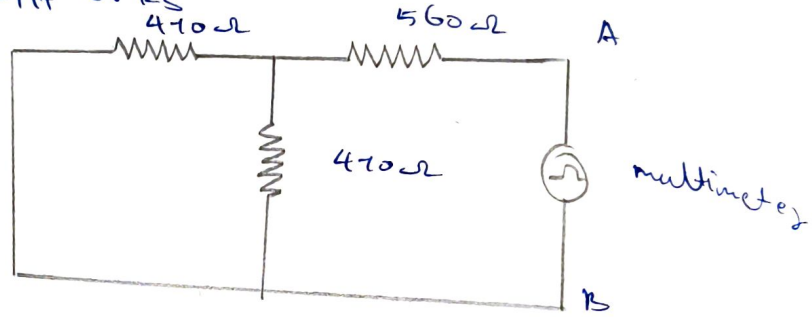
Circuit Diagram: verification of thevenin's theorem

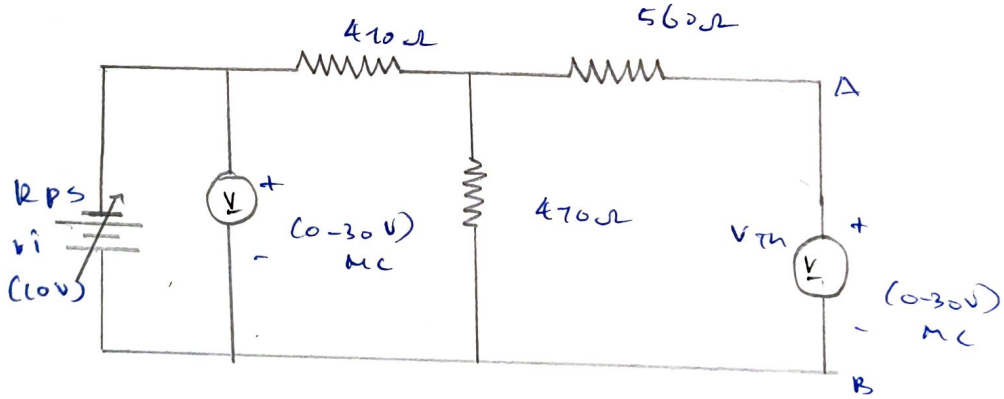
To measure $I_L$.



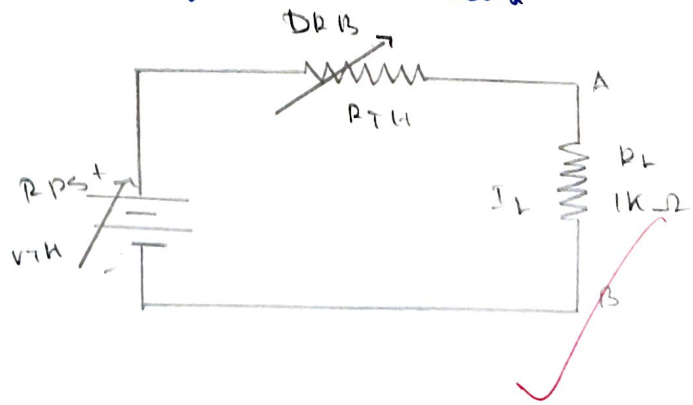

To measure $R_{TH}$ or $R_S$




To measure $V_{TH}$ or $V_{OC}$




Thavenin equivalent circuit



Exp. No: 4

Date: 04/02/2026

# Verification of thevenin's theorem

Aim:

* To verify thevenin's theorem and theoretically for the given DC circuit.

Apparatus Required:

| Sl. No | Apparatus | Specification | Quantity |
|---|---|---|---|
| 1. | Regulated power supply (RPS) | (0-30V) | 1 |
| 2. | Voltmeter | (0-30V) MC | 1 |
| 3. | Ammeter | (0-10mA) MC | 1 |
| 4 | Resisters | 470Ω | 2 |
| | | 560Ω | 1 |
| | | 1KΩ | 1 |
| 5 | Bread board | – | 1 |
| 6 | multimeter | – | 1 |

Theory:

Thevenin's theorem:

* Thevenin's theorem states that a linear terminal circuit can be replaced by an equivalent circuit of a voltage source $V_{TH}$ ($= V_{OC}$) in series with a resistor $R_{TH}$. where $V_{TH}$ is the open circuit voltage at the load.

Tabular column: 1

To measure $I_L$

| $V_i$ (volts) | $I_L$ (amps) |
|---|---|
| 10V | 2.8 mA |

Tabular column: 2

| $R_{TH}$ ($\Omega$) | $R_N$ ($\Omega$) |
|---|---|
| 745 $\Omega$ | 745 $\Omega$ |

| $V_i$ | $V_{TH}$ (volts) |
|---|---|
| 10V | 5V |

Calculation:

To find $V_{TH}$.

By KCL

$$\frac{V_{TH} - 10}{470} + \frac{V_{TH}}{470} = 0$$

$$2V_{TH} - 10 = 0$$

$$V_{TH} = 5V$$

By KVL

$$10 = 470I + 470I$$

$$10 = 940I$$

$$I = 0.010A$$

$$V_{TH} = IR$$

$$= 0.010 \times 470$$

$$V_{TH} = 4.98$$

$$V_{TH} = 5V$$

Thevenin's voltage, $V_{Th}$:

* It is the open-circuit voltage across the Load terminals (AB). i.e $V_{TH} = V_{OC}$

Thevenin's Resistance, $R_{Th}$:

* It is the Look-back resistance across the Load terminals (AB). i.e $V_{TH} = V_{OC}$

Thevenin's Resistance, $R_{Th}$:

* It is the Look-back resistance across the Load terminals when all the sources are replaced by their internal resistance An ideal voltage source is replaced by short-circuiting its internal resistance is zero. An ideal current source is replaced by open-circuiting as its internal resistance is infinity.

Procedure:

* make the connections as per the circuit Diagram: 1

* vary the RPS and set an input voltage of 10V.

* Note down the voltmeter reading ($V_i$) and ammeter reading ($I_L$) in Tabular column 1.

* Switch off the Supply and make connections for circuit Diagram 2.

* measure the thevenin's resistance $R_{TH}$ = Norton's resistance $R_N$.

To find $R_{TH}$:

$$R_{TH} = \frac{470 \times 470}{470 + 470} + 560$$

$$R_{TH} = 795\,\Omega$$

* calculate the $I_L$ value using the formula

thevenin's theorem

$$I_L = V_{th} / (R_{Th} + R_L)$$

Norton's theorem

$$I_L = I_N * R_N / (R_N + R_L)$$

* theoretically verify the Norton's theorem

mark split-up:

| SL.NO | Description | mark allotted | marks obtained |
|---|---|---|---|
| 1. | Experimental setup | 20 | 20 |
| 2. | observation / reading | 20 | 20 |
| 12. | calculation / results | 20 | 20 |
| 13. | viva | 20 | 17 |
| 14. | Record | 20 | 18 |
| | | | 95 |

Result:

* thus. thevenin's theorem is verified practically and Theoretically.